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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Existence of Cohen rings with prescribed residue field and the Cohen structure theorem. Algebra, Definition \ref{algebra-definition-cohen-ring}, Algebra, Lemma \ref{algebra-lemma-cohen-rings-exist}, and Algebra, Theorem \ref{algebra-theorem-cohen-structure-theorem}, cited at proof lines 9828-9837. Provide the power series presentation of the common quotient and the characteristic split.
- Power series rings over a Cohen ring are formally smooth, and continuous maps lift along surjections of complete local rings. Lemma \ref{lemma-power-series-ring-over-Cohen-fs} and Lemma \ref{lemma-lift-continuous}, cited at proof lines 9840-9842. Lift the surjections onto the common quotient to maps from the power series ring.

Source reference. Original lemma 09Q8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12276–12276. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0327: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4010–4010. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0328: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4011–4011. Navigation pointer to a statement the

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Source reference. Original lemma 032A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4013–4013. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9766–9766. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9764–9764. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $S \to R$ and $S' \to R$ be surjective maps of complete Noetherian local rings. Then $S \times_R S'$ is a complete Noetherian local ring.*

Proof. Let k be the residue field of R. If the characteristic of k is $p > 0$, then we denote Λ a Cohen ring (Algebra, Definition 0327) with residue field k (Algebra, Lemma 0328). If the characteristic of k is 0 we set $\Lambda = k$. Choose a surjection $\Lambda[[x_1, \ldots, x_n]] \to R$ (as in the Cohen structure theorem, see Algebra, Theorem 032A) and lift this to maps $\Lambda[[x_1, \ldots, x_n]] \to S$ and $\varphi : \Lambda[[x_1, \ldots, x_n]] \to S$ and $\varphi' : \Lambda[[x_1, \ldots, x_n]] \to S'$ using Lemmas 07NL and 07NJ. Next, choose $f_1, \ldots, f_m \in S$ generating the kernel of $S \to R$ and $f'_1, \ldots, f'_{m'} \in S'$ generating the kernel of $S' \to R$. Then the map

$$\Lambda[[x_1, \ldots, x_n, y_1, \ldots, y_m, z_1, \ldots, z_{m'}]] \longrightarrow S \times_R S,$$

which sends x_i to $(\varphi(x_i), \varphi'(x_i))$ and y_j to $(f_j, 0)$ and $z_{j'}$ to $(0, f'_j)$ is surjective. Thus $S \times_R S'$ is a quotient of a complete local ring, whence complete. □